

K4—Companion

MacOS Python Installation

Please read and follow all steps carefully. These instructions are verified on several MacOS machines running MacOS 15.5 and 26.5. It should work on other versions of MacOS as well. The usual disclaimers apply. Install at your own risk. Management, writers, producers, programmers, and all others associates are not responsible if you break your computer. Having said that, there is little to be broken and there are no reports of any problems. This document assumes knowledge beyond that of a typical “appliance user.”

1. Install Xcode from the App Store. After downloading, open Xcode. There is another Download and Install button. Select MacOS, not iPad or other versions as they are not needed. It does not actually download the code again but it does install the necessary code. Accept the license for Xcode. After completely installing Xcode, open a Terminal window (Applications → Utilities → Terminal). Run: `sudo xcodebuild -license` and agree to the license terms. It has been found that MacOS does not always require this step
2. Download and install the latest version of Python 3 from <https://www.python.org/downloads> Python 3.13 or later is needed: the encrypted connection to the K4, on port 9204, requires it. Use the “macOS 64-bit universal2 installer”
3. Optional step to use the Bash terminal shell. If you prefer to use zsh, the default Terminal shell, see below. Change the default Terminal shell to Bash. Open a terminal shell (found in Applications → Utilities). Run: `chsh -s /bin/bash` This step is due to Bash being commonly used in many Linux/Unix applications. Again, zsh shell users see comments near the end of this document
4. Create an alias to the current version of Python. The default version of Python3 installed on macOS is 3.9.6. Creating an alias points to the version just installed. In Terminal, run: `nano .bash_profile` Note the ‘dot’ at the beginning of the filename. Enter the following in `.bash_profile`: `alias python=/usr/local/bin/python3`. On a second line, enter: `alias pip=/Library/Frameworks/Python.framework/Versions/3.13/bin/pip3`, with 3.13 replaced by the version installed, for example 3.14. Note the actual commands do not use the single quotes ‘’. Issue the command `source .bash_profile` <enter> to activate the changes. It is recommended to completely close Terminal (from the menu bar) and re-open, especially if the new ‘alias’ entries are not recognized. This will re-read the configurations and system file locations. Zsh shell users see below for instructions.
5. Next, type: `pip install numpy` <enter>
6. Opus codec is next dependency. This is a download, build, install process. It is not at all complicated. Go to <https://opus-codec.org/downloads/> and download the file `opus-1.5.2.tar.gz` This will likely go to the Downloads folder on your Mac. In Terminal, type `tar zxvf opus-1.5.2.tar.gz` <enter>. This will unpack the Opus library build files into a directory `opus-1.5.2`. In Terminal, type the following commands: `cd opus-1.5.2` <enter> then `./configure` <enter> (and note the ‘dot’ before the ‘slash’). Next type

make <enter> and finally *sudo make install* <enter>. You will be asked for your password at this point, and the files will install into their proper places.

7. Next in Terminal, type *pip install opuslib* <enter>
8. Next, install the remaining Python packages: *pip install sounddevice pysocks pyyaml PyQt6 pyqtgraph python-rtmidi pyserial* <enter>. *sounddevice* brings its own copy of the PortAudio audio library, so PortAudio does not have to be built. *PyQt6* and *pyqtgraph* draw the user interface and the panadapter, and *python-rtmidi* and *pyserial* connect MIDI and serial keying devices.
9. Install MacPorts <https://guide.macports.org/chunked/installing.macports.html> . Download the appropriate version and click on the package to install. You should once again completely exit Terminal, then restart to access '*port*'
10. Install *pkgconfig* with Terminal, using the command *sudo port install pkgconfig* <enter> Allow the install to add the needed dependencies. Install *cmake*: *sudo port install cmake* <enter> Allow all dependencies to be installed 'Y'. Install *cairo*: *sudo port install cairo* <enter>. Allow all dependencies to be installed 'Y'.
11. Install Python *pkgconfig* *pip install pkgconfig* <enter>
12. Install *pyobjc* *pip install pyobjc* <enter>

If all has gone well, download *k4companion* from W7DA's repository on github at <https://github.com/DaleFarnsworth/K4-Companion> Warning: if downloading from the web pages, the file *k4companion* will download as *k4companion.txt*. This file should be renamed as *k4companion* with the file extension deleted. To run the application directly, go to Terminal. Navigate to wherever you have placed the K4 Companion files. Type: *chmod 755 k4companion* <enter>. A *k4companion.yaml* file is no longer needed: the configuration is built into *k4companion*. With this installation of K4 Companion, the program must be started from the command line. Again, navigate to the folder (directory) containing *k4companion*. To start the program, use the following command *./k4companion &* <enter>. The ampersand '&' forks the program into the background. Then the terminal window can be closed. You should be ready to put the radio's information in the Server tab at this time. K4 Companion connects on the K4's encrypted port, 9204, so no port needs to be entered. Since MacOS has extra security, the first time opening the program you must "Allow" K4 Companion through macOS security. Because it couldn't connect immediately, it is necessary to press "Connect" after allowing the program. If you are familiar with shell scripts, there is a way to create a desktop or other shortcut that can be called from Finder. When the script is run for the first time, it will advise of any missing dependencies or skipped steps.

Note for zsh shell users:

To create an alias with zsh, open Terminal. You should be in your home directory (type *cd* <enter> if you have navigated elsewhere. Type: *nano .zshrc* The alias will take the following forms: *alias python="/usr/local/bin/python3"* and *alias pip="/Library/Frameworks/Python.framework/Versions/3.13/bin/pip3"* With zsh the double quotes are needed. It is best to close Terminal and reopen so paths can be properly established each time changes are made and before MacPorts is installed.

It is possible to create a Mac app from the raw Python script, if you prefer to launch from an icon or from Applications. Below is the process and a sample script. First, obtain pyinstaller `pip install pyinstaller`. Then run the command:

```
pyinstaller -w --noconsole -i k4.icns k4companion
```

Omit `-i` and `k4.icns` if no icon is available.

Creating a dmg is a little more involved. The source for 'create-dmg' is available on Github.

```
create-dmg \
--volname "K4-Companion Installer" \
--volicon "/Users/$USER/bin/dmgicon/k4-logo.icns" \
--app-drop-link 12 12 \
--no-internet-enable \
dmg/K4-Companion.dmg dist && swift /Users/$USER/bin/dmgicon/seticon \
/Users/$USER/bin/dmgicon/k4-logo.icns \
/Users/$USER/bin/dmg/K4-Companion.dmg
```

These processes can be combined in a single script:

```
#!/bin/sh rm -rf dmg/K4-Companion.dmg
rm -rf dist/k4companion* pyinstaller -w --noconsole -i k4.icns k4companion
rm -rf dist/k4companion
create-dmg \
--volname "K4-Companion Installer" \
--volicon "/Users/$USER/bin/dmgicon/k4-logo.icns" \
--app-drop-link 12 12 \
--no-internet-enable \
dmg/K4-Companion.dmg dist && swift /Users/$USER/bin/dmgicon/seticon \
/Users/$USER/bin/dmgicon/k4-logo.icns \
/Users/$USER/bin/dmg/K4-Companion.dmg
echo ""
echo "The process is now complete."
echo ""
echo "The K4-Companion installer is found in /Users/$USER/bin/dmg/"
echo ""
#EOF
```

Of course, the script must be modified for your file locations, icons, and general setup. An App store version of K4-Companion will not be made available, due to Apple licensing requirements.

Note for MacOS users: RUMlogNG is a free (and excellent!) logging utility for MacOS, and the developer is Thomas – DL2RUM. The two programs run perfectly side-by-side. Connect RUMlogNG via TCP/IP by using the radio's address and port 9200 with the radio selection as Elecraft K4 – ctrl. RUMlogNG is available free from the App Store. RUMlogNG configuration page.

Settings for running side-by-side with K4 Companion

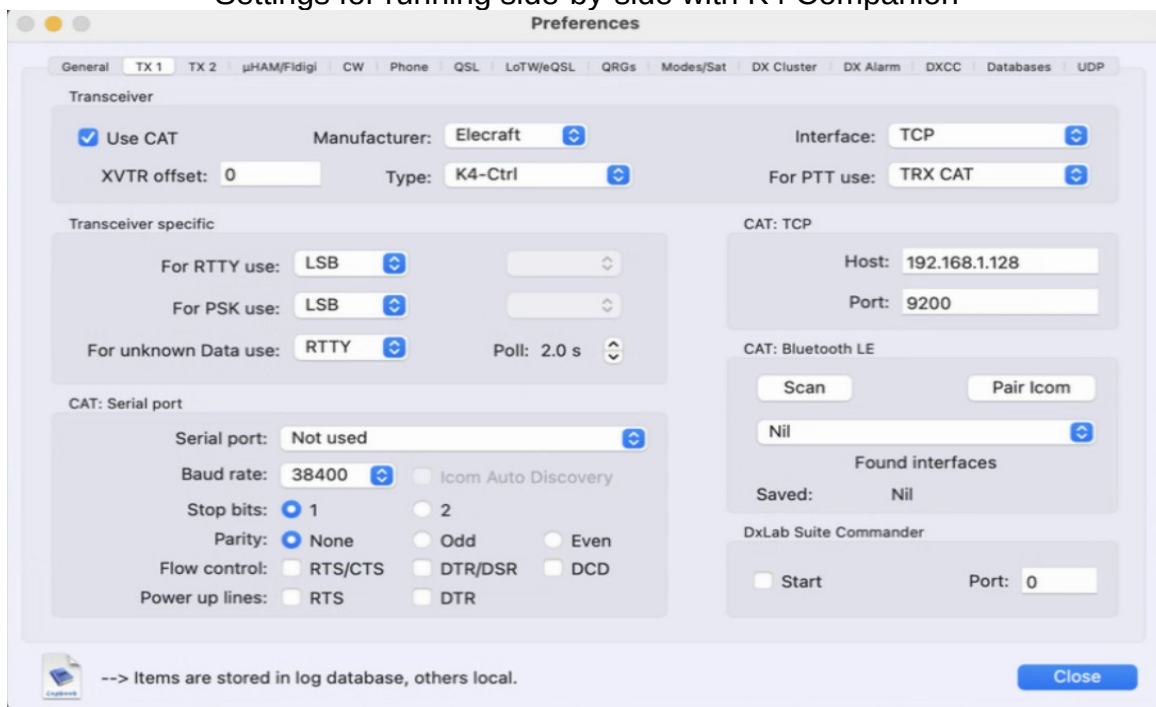


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